

LSTM Architecture

Deep Learning

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Contents of This Video

In this video, we will cover:

- The LSTM cell architecture
- Forget gate: what to discard
- Input gate: what to store
- Output gate: what to output
- The cell state: highway for gradients

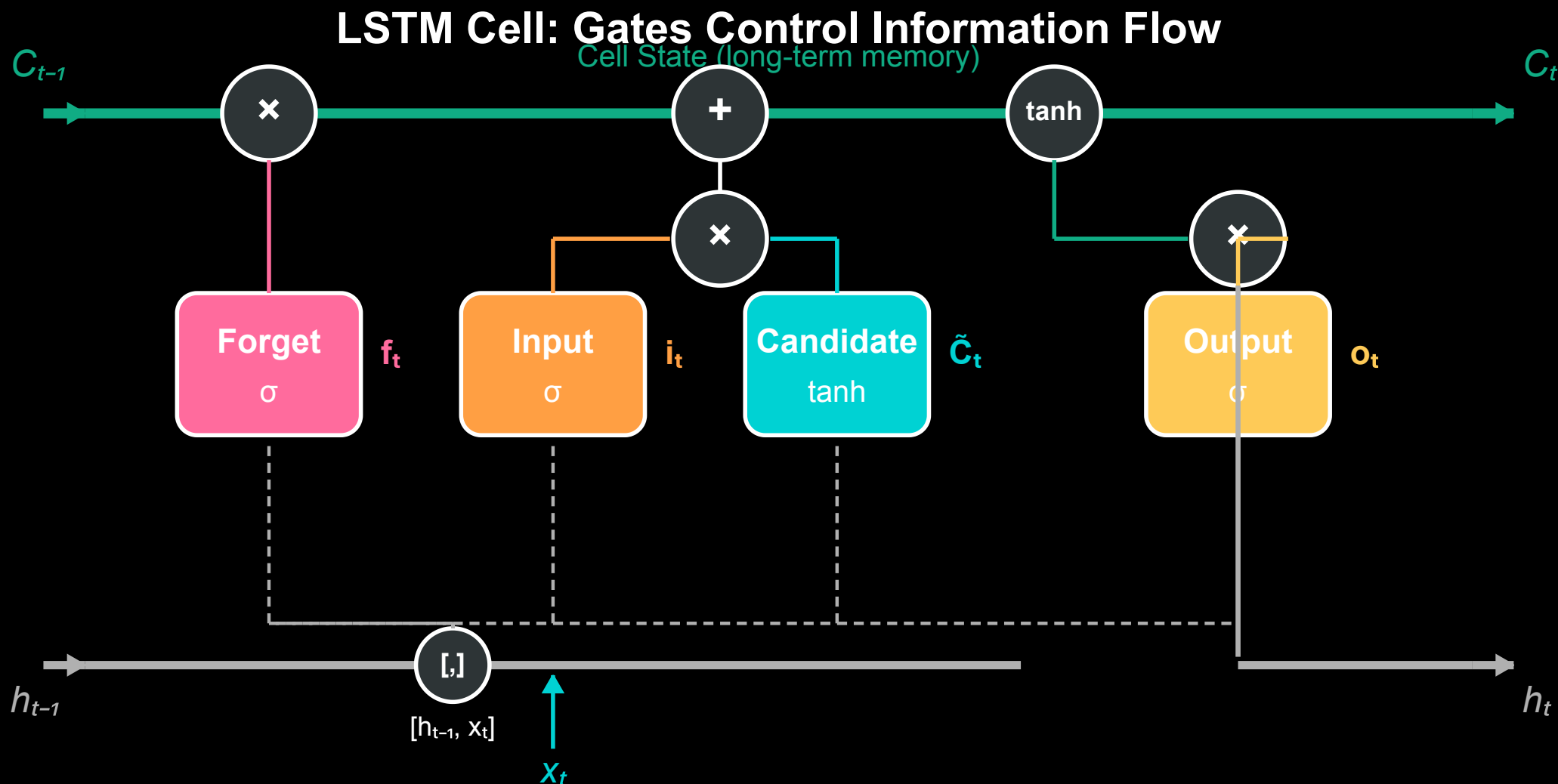
The Core Idea: Gated Updates

Basic RNN problem: Every timestep completely updates hidden state

LSTM solution:

- Separate “cell state” for long-term memory
- Gates decide what to keep/add/output
- Cell state can pass through unchanged

LSTM Cell Overview



Forget Gate

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

What it does:

- Looks at previous hidden state and current input
- Outputs values between 0 and 1 for each cell state dimension
- 0 = “forget this completely”
- 1 = “keep this entirely”

Example: When starting a new sentence, forget old subject

Input Gate and Candidate

Input gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

Candidate values:

$$C_t^{\sim} = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

What to add to cell state:

$$\text{New info} = i_t \odot C_t^{\sim}$$

The input gate scales how much of each candidate value to add

Updating the Cell State

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

In words:

New cell state = (Forget some old) + (Add some new)

Key insight: Addition lets gradients flow back unchanged!

- No repeated multiplication through the same matrix
- Gradients can travel many timesteps without vanishing

Output Gate

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(C_t)$$

What it does:

- Cell state contains the memory
- Output gate filters what to expose as hidden state
- Hidden state is used for predictions

LSTM: Full Equations

Forget gate: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$

Input gate: $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$

Candidate: $\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$

Cell state: $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$

Output gate: $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$

Hidden state: $h_t = o_t \odot \tanh(C_t)$

What We've Covered

In this video, we've learned:

- LSTM uses gates to control information flow
- Forget gate: decides what to remove from memory
- Input gate: decides what new info to store
- Cell state: long-term memory with additive updates
- Output gate: decides what to output
- Additive updates prevent vanishing gradients